Karl Friston, Adam Goldstein, and Michael Levin discuss active inference and algorithms

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great so uh what I was hoping Carl um is that we could get your thoughts on uh how the whole uh active inference framework could be applied to something that uh we've been developing and uh I don't know if if you had a chance to look at any of this stuff but I'll just give you a brief uh you know kind of a brief summary so that it's clear what what we've got and then and then I have some basic questions and then a couple of wacky ideas to to kind of bounce off of you and see what see what you think so the the basic thing is this that what what what I was trying to do is to have a very uh basil model of um distributed uh intelligence and the idea was that uh we were interested in um unexpected competencies in places where unlike in biology you know in biology no matter how simple your model you never have all the information about mechanisms and somebody can always say well there is a mechanism for that you just haven't found it yet right so we wanted something that was incredibly uh simple incredibly transparent uh deterministic something that everybody thinks they know what it does and then we can we can apply some of the approaches that we take in my lab about taking something that doesn't seem cognitive and saying okay but what what actual competencies might it have right and so we chose this thing called sorting algorithms and so these are the same uh simple algorithms that all computer science students study for you know and they've been studied for decades and then we made a couple of uh a couple of twists to it is that we visualize their progress from being a jumbled set of digits to an ordered set of digits as a kind of a traversal of of space right so the idea is they start in different locations and they all sooner or later they all end up in in one location where everything is and so once once you view it as as as navigating that space then you can ask some questions about what are their competencies in navigating that space under under odd perturbations one of the perturbations that we made was the um introduction of what we call broken cells or or barriers in the space so if the algorithm wants to swap two numbers in order to proceed in its in its sorting trajectory well one of the numbers could be broken it doesn't move you can't you can't move and so and and we have two kinds of broken numbers ones that um never initiate swaps and ones that actually never uh never swap no matter you know who initiates them and so that allows us to ask uh questions about uh things like delayed gratification in other words can it go further from it if if it encounters a barrier can it go further away from its goal in order to then acquire gains afterwards and this is you know William James talked about this of course as H as an important type of Basil intelligence and um and that and that breaks a common assumption with these algorithms which is typically you assume that the material is is um robust in other words when you when the algorithm says to to do something it gets done but but in our case not necessarily and we never introduced this is important we never introduced any extra code to check if things got done it's it's the standard algorithm so it just keeps on rolling there is no code to see how am I doing did things work out no no code for any of that the second thing we did was to um break the uh the the the general version of this is centralized so there's like this omnicient controller and it's following one of several algorithms to kind of move numbers around and we got rid of that and instead made it all bottom up so every digit AKA every cell now has its own uh version of the algorithm running and it has a a li local view of who its neighbors are and it's just following the steps of the algorithm to try to improve its local environment that there is no more global global control so it's distributed and uh and and you know we learned a few things we learned that a the distributed version of this works quite well um it actually you know they do sord nicely so so that's great we did see some delayed gratification in the sense that if you if you sprinkle in some broken cells it actually will go backwards and unsort the string a little bit and then effort to then go around the the defect as it were so that's kind of cool but the most kind of surprising thing which is what um I'd love to get your your take on is this um we because now it's distributed and every cell is following its own algorithm that enables us to do an experiment that otherwise you couldn't do which is to make a chimeric string and it's what we do in developmental biology when we put together you know axotal cells and frog cells and now you get this frog aot and you can ask questions like well what shape is it going to have right so so what you can make is a chimeric string where some of the numbers are following one algorithm some of them are following a different algorithm and again there is this is important there is no code to uh determine either your own or your neighbor's algot type and algot type is a word that Adam cone for this like um you know what algorithm what what set of properties are you actually following a policies are you actually following so there is no code for any of that but we know um which which algot type all the cells are so that and and

and that also works okay Chic strings also sort sort them sort the arrays and that's fine what we then did was we asked basically a developmental biology question was to say okay at any particular point during its Journey what is the distribution of algot types within the string and what we know and and we Define of a quantity called um clustering which basically just means uh you look you look next to you and what what's the probability that the your neighbor next to you is the same algot type as you are so what happens is that in the very beginning that probability is 50% because the algot types are randomly assigned to the digits so 50% that's our Baseline at the very end it's also 50% because at the end everybody has to be sorted and there is no relationship between the actual sort order of the numbers in the algot type so again it's 50% but the wild thing is that in between those two points if you actually plot that curve over time it actually goes like this and in between it's quite a bit higher even if it's statistically very significantly higher than 50% and what we see is clustering significant um tendency of cells with the same algot type to locate close together eventually the the the inevitable the inevitable physics of the algorithm will yank them apart and make sure that everybody's in numerical order but until that happens they enjoy some amount of clustering with their you know with their cons specific so to speak until then and so you know any thoughts you might have but more specifically like one hypothesis the one could make even though there's no explicit mechanism for this but it might be a you know an emergent thing could they be preferring to be next to their neighbors because their neighbors be be ones of the same algot type are more predictable right it's it's less surprised you're less surprised when you're sitting next to somebody who's following exactly the same policies as you are so so I'm curious what you think about that and I'm curious um if there you know what what might be uh a set of experiments that we could do to test that that what's going on here is some sort of implicit surprise prize minimization even though there's no actual code for it so I'll stop there and and listen to what you've got to say sorry um I think you're still muted I was saying that was a very succinct and clear and nice summary I I reread the paper a couple of days ago just to refresh s um my memory for this conversation um so I didn't realize that that was the um that final cam um demonstration was U sort of you know the most intriguing from your point of view um but indeed the way you express it you that that does call for um um further analysis understanding and numerical experiments um so overall um just to um endorse the choice of the Sorting algorithm as if you like a minimal um kind of of uh self-organization um I think that you know that the self-organization word needs to be um center stage in terms of uh you know what you're trying to understand here um and framing like that it does remind me a lot of self-organizing maps I don't know if you remember Vander murgs um uh treatment and people who's this Peter Dian supervisor I've forgotten now um so this notion of self-organizing maps as a very sort of btic aspect of self-organization I think um certainly puts sorting like algorithms um um center stage in terms of biological self-organization particularly in things like the visual cortex and why you get um that kind of um pin wheel architecture for example where where receptive field properties tend to cust together in a smooth way and you get all sorts of interesting symmetry breaking when you're trying to represent say a 5D perceptual space on a 2d manifold so that does strike me having this sort linear linear sorting algorithm um and without really having not thought about it before but certainly the the the Dual um the Dual pressure to find a free energy minimizing solution viewing free energy as an extensive quantity in more simply you know the collective free energy being the joint free energy minimum solution you're asking your where the free energy dis bounds the likelihood of um this particular Arrangement um then you're looking for um the precise functional form of the free energy um and um if you've got this kind of um opponency between the similarity of the algorithm and the similarity of the content the value then I can certainly see interesting behaviors arrive arise in exactly the same spirit that you get these interesting um structures in not epithelia but in in the functional specialization of of cortical representations of sensory epithelia um that try to sort of pack what you three dimensions into one dimension or are accountable to two kinds of constraints so I'm you know again without really thinking about it because I wasn't I wasn't anticipating that particular question but I think what you would be um the first thing that you would be looking for is basically what is the lran or the energy function that is being minimized so you could regard this as um the Sorting algorithm as a um or an application of the Sorting algorithm as a process um that is trying to minimize some energy function very much in the spirit of Mark of random Fields but in your instance you've just got a one-dimensional field but this the the the technology of markof random Fields I think would be apt to try to understand the functional forms of the energy functions um where under the special constraint which of course is the one that you're predicating this whole thesis on the interactions are only local and therefore any um Collective Behavior has to be an emergent property which is truly distributed so the definitive aspect of a Markoff random field is you just have local just have local interactions and that I think that's a really another important architectural feature that comes along with the choice of the Sorting algorithm which you should foreground um because you know any distributed Collective um or emergent Behavior at a scale Beyond local interactions that emerges is truly emergent in the sense that your all your all your interactions are local and of course that is what the markof field random field gives you it says that you can only express the excuse me the energy function which is the probability of getting this particular Arrangement or these particular numbers um in this local cque uh you can only express that in terms of a local energy function and then of course you could tell sorts of stories about the importance of that for machine learning and the like um but you you you probably want to stick to self-organization so um yeah and I would imagine the energy function is now going to be some simple um um measure of the um the local differences or the local gradients uh and of course what you know what one would anticipate would be a smoothing um a resolution of as you say the the other the differences um so I think that would be one way of approaching um you know naturalizing this phenomena in terms of math by just invoking a um an arbit not a variational free energy in the spirit of the free energy principle but just a A lran or try to identify what is the uh the generic free energy functional that's being minimized here um so that now your view through this um sorting space or your morphological space um is now a progression on some wadington landscape that is defined by the um this free energy functional um and whether you can reverse engineer that or not I you know I don't think it really matters other than because the I think the nice aspect of that is then you can talk about the Dynamics on this free energy landscape um and one's LED then to very similar sort of Notions in computational chemistry and protein folding and the like you know that there's a very complex um sorry there is a complex Wallington landscape or free energy landscape that self-organization and computational chemistry um aderes to and can be understood in terms of freey Minima indeed most of computational chemistry sort of follows this um and indeed that is identifying that landscape is the whole point of applying things like large language models or um deep RL to to sort of protein folding and um other applications um so that would that that would be certainly one view to get a free energy like um formalism or naturalization of of this Behavior which I I repeat has lots of really interesting links with with um self-organizing Matt's Mark or random Fields image reconstruction and self-organization in um you know certainly in things like the visual cortex I would imagine any mapped representation would would would conform to to uh to to these rules um to get this into a to get it into um a free energy principle uh um story um I think you would you you'd have to commit to the notion that each of the cell has its own boundary and now you're starting to interpret each number as a thing and in so doing a acknowledge its openness to everything else or in this instance just its neighbors um uh which would require sort of a bir your formalism of the bidirectional exchange so that the value of my nextdoor neighbor is something that I can sense and you and is uh and likewise um the broadcasting of my number to the next door neighbor is something is an action so you've got this um openness that is mediated in the simplest way which is just the broadcasting and sensing of one unid dimensional one number in fact it's a discrete number um um and viewed like that that means you can then I think deploy the free energy principle um in the sense that any uh non-equilibrium or um far from equilibrium steady state which I think you would probably have here just in virtue of the fact that there is going to be some um breaking of detail balance in the in the itinerant way in which you move through this space in a developmental to get to your to your steady state um and one can imagine well perhaps not but you know if one puts a little bit of Dynamics into this that I would imagine you would evince very very clearly the breaking of of detailed balance um and and you know and have you know those kinds of solenoidal flows um I mean sorry I I distracted myself just by um the addition of the Frozen cells that's one way of breaking um in a sense uh the detailed balance and you just Open brackets it's exactly the same device that um I resol reported to in the very first paper um um on the uh Life as We Know It paper when simulating the little um macro molecules using Loren attractors that had inherent Dynamics but to make it interesting you had to have a small a certain popul number of the of the synthetic macro molecules that were insensitive to influences from other macro molecules and another proportion that could could not influence the other one so it's almost exactly the same Choice uh and that's what gave it the interesting Behavior otherwise it just basically converged either to um a gas or it uh at a certain temperature it would just converged to a um to a crystal um so you both of them being steady state solutions for energy minimizing Solutions but things got interesting when you broke the detailed balance your symmetry breaking by having this um this you know this requisite Variety in terms of um the the frozenness in terms of action or or sensation so I think that's another that's another important thing to foreground that you know this may be this this kind of requisite variety may may be absolutely necessary for symmetry breaking and in this particular instance breaking detail bounds to get um this kind of self your know bi biological Ally plausible or btic kind of self organization you're unlikely to get that if um you you you're in the absence of it in the sense that it would um converge to a crystal in your instance just a linear sorting perfect sorting which which is which is you know just doesn't have that chimeric or itinerant I itinerant aspect to it so um sorry Clos brackets wait oh yeah so if you got now an interesting system that has a non-equilibrium steady state and in your case actually because you haven't got Dynamics um it will also be an equilibrium steady state but it'll still be a free energy minimizing solution um then you are perfectly entitled to interpret the numbers as things uh and inferring things uh and all they're trying to infer is the cause of their Sensations which is just the value of the numbers on one side and the other side and they are broadcasting their inferences through um um um broadcasting their own number um which of course will be the average of um well when sorted it will be the average of uh the the neighboring numbers so on that view I think you could very easily license an active inference interpretation a thology you you I mean you wouldn't actually need this to simulate uh protein folding or self organizing Maps or anything but you would be able to say there is a a teic interpretation of the self-organization using the rhetoric of inference and belief updating simply because we can treat each number now as a Markoff blanket and the then something which will never be accessible but we can um imply or induce um internal to each number um would um uh could be interpreted as an inference process and then the story which you you've already said what the the answer is that um you know under the assumption that um I live in a world that is um maximally predictable um then everything around me is the same as me and therefore um um uh I am going my free energy my variational free energy um minimizer is now going to be um found when there's the least surprising input and if I believe that everything is like me um then that will be when the numbers um that I am sensing in my click are as similar to my uh the estimate of the number um um that um the place that I should be coming back to our sort of knowing your place paper um and um I think that that that that that kind of story um will have to be nuanced for the same algorithm um so you know I'd have to think about that a little bit more but certainly at least at a narrative level or a conceptual level I think you can tell the same story there that if the sequence of moves that I see my neighbor doing in relation to what I know about my neighbor belies the same underlying dynamic or algorith your algorithmic um computations then in some sense they are predictable if I have exactly the same algorithm under the hood and therefore um mathematically speaking um that should be um that would be the free energy minimizing solution if I can now read my broadcasting of the number as a broadcasting my posterior beliefs about you know the kind the number that I the estimate of this local you know my Niche um um uh in this instance is just labeled with one you know with one with one number so that the the number that I have um is basically um my PRI belief about my Niche and I'm just now going to um move my Niche around in a sort of you know egocentric frame uh until it is consistent with my Prim belief that this is my place my Niche is number 62 for example um and that should be you should be able to reproduce the same kind of sorting um either analytically through showing that um with an appropriately configured lran or free energy functional that this the system operationally is appears to be minimizing um you can um you can now write down the generative model and then show that this can also be interpreted as an influence process I repeat under the assumption that um the best way to make the world predictable is to surround yourself with things like you um um which and also of course the locality assumption that I can only talk to the person to whom I'm uh immediately connected so those are some of my thoughts but a lot of those were invented on the fly in response to your question I'm afraid superb um I've got many questions but Adam why don't you ask yours yeah so so it strikes me that up until now we've talked about the relevant sort of agent as being the individual number with an algot type you can think of it as a cell but it strikes me that there's an interesting macro phenomenon that that occurs in the process of sorting which is that it appears that the that the list actually minimizes the kov complexity or the description length necessary to render it right so let's just say you've got an unsorted list with random you know random distribution of algot types and there's 10 items in the list you would need to enumerate 10 numbers and 10 algot types and there's no reason a priority to think that that would be compressible in any way I mean maybe maybe you'd get lucky and there'd be a string of a certain number a string of a certain algor type but in the general case I think you'd actually need to write out every single entry but as the list starts to sort itself it actually starts to create these longer strings of algot types which means that the minimum description length actually gets shorter yeah you still need to write each number out but you can core screen the descriptions of the algot types you can say the first five numbers have the same you know algot type and then the next three have the same and so on now that that's a macro phenomenon but I'm wondering if there's any any evidence or any research that suggests that sort of these self-organizing systems have a tendency to minimize their description length to minimize the number of factors needed or something like that because if that's the case then it gives us another view where there's a sort of emergent complexity minimization happening at the at the collective level yes no that's an excellent point um I think you know the simple answer is yes absolutely um uh and I can I can sort of give you my take on the literature or the citations that you'd want to appeal to but I I should say it's going to be a nuanced yes because of the particular focus on the clustering of the algo type now the algo type induces a certain kind of Dynamics into the game so it's not as simple as a self-organizing map it's how the map actually self-organizes so there are a certain sort of uh there's a process under you know under under the hood um and that I think makes it slightly more complicated than just understanding self-organized Maps but in terms of um in terms of um another thing you might want to look into here of course and you probably know more about this than I do but you know this has um a lot of resonance with artificial life um g games in the 1990s 1980s it also um could be um um if you want to so do um interest inly linked to Steven Wolf's ruad which is Al Another local um scheme um that generates everything apparently uh so there's the same sort of notion so he has algorithms which he calls rules and the rules are recursively applied in a local fashion to generate everything um including black holes apparently and and quantum physics and everything uh so you and just you might there might be an interesting point of contact here um with with these uh sorry but to come back to the simple answer yes absolutely so um certainly from the point of view of self-organization as described by the free energy principle so notice here um the free energy principle is just a description of systems that self-organized to a far from equilibrium non- equilibrium steady state it's not a recipe for sorry in its statement it is not um a a description of a theological description of inferential uh processing you are licensed to to equip your explanation of this self-organization with reference to inference but that's an application of of the free energy principle in itself it's just a description of any anything that self-organizes or any things that self-organize um so in that sense um you know if there is self-organization under the could then the free energy principle has to apply and you can motivate the free energy principle um along two lines one would be the sort of playing the Fineman card which is basically um derive um looking at the um the minimization of uh free energy as uh um an optimization process uh which can be viewed as a gradient descent on some Fitness landscape or fre ape or Wasington landscape um or you can take the Russian perspective which would be the col of complexity and from the com Comm complexity you you you get to uh solof induction and from that you get to Universal computation which is the home of the minimum description length and minim minimum message length so it's the algorithmic complexity version of free energy and so David McCay um wrote a quirky little paper I think 1992 of of four where he interpreted variational free energy in relation to um minimum message length um using crypto analysis as a vehicle to tell that story but to my mind I think wonderfully connected two different two different perspectives on exactly the same phenomena the you know ways of describing self-organizing systems um that basically both entail a minimization of complexity um a simplification an emergence of order of of a particular sword that entails um um um either compression um hence the minimum description or the minimum message length uh view from the algorithmic complexity in terms of um sort of you know rate coding theorems rate Distortion theorems and the like um or you can write it down in terms of continuous probability distributions and and sort of follow through from the finan path integral I think they're both saying the same thing you know you know the way I think of this is that the you know the end point of any self-organizing thing or set of things is just going to be the most likely configuration that they occupy given the kind of things that they are um and that basically um means that you can always describe this um in a statistical and Theological sense as um everything um providing an accurate prediction of what it senses that is minimally complex in exactly the same Spirit as um the um the way that you would frame complexity in terms of um loss or not loss lossy compression or minimum description length or minimum um algorithmic comp complexity so um I think that if you if you um if you tell a story that the free energy is an extensive quantity which means that all the set of numbers or any subset of numbers any partition um will all will all look as if they are minimizing a free a free energy functional um then you can I think um say that you know one view of this functional is to minimize the complexity of the arrangement uh which should be manifested in terms of a minimization of algorithmic complexity and you can use is it I never I can never ZL lip or lip lymph whatever do you know what I'm talking about was one of these um um hierarchical sequential entropy measures um um you know there one way of quickly enumerating the um the algorithmic complexity so I think the you if you could join the dots that would be a really really powerful um view of this and indeed you know it would be interesting if you could um just using numerical experiments join the dots um quantitatively uh in terms of this handcrafted inted uh free energy lran just based upon you given three numbers you have to now write down an energy function that is always going to be minimized the Sorting algorithm so that the end point um conforms shares the same Minima of your energy function it could be really simple it could be this the two differences squared and added together something as simple as that um and if you can prove that the the Minima of this is the same um uh is is is the same as the um the end point of your self-organization then you say this is one fre energy functional the very much in the spirit of hot field Nets and Harmony functions you know in the early days of um uh neural networks um spin glass models uh pots models all of these um Mark of random Fields uh you have to write down this kind of energy function and then you just simulate you know you can you can just do a grad in descent or rearrangement in order to minimize that um so that's one very simple kind of free energy um description of it then you would have um an inferential one under an assumed generative model so if you assume each number actually has a little mind and a gened model it's trying to estimate or trying to act upon its world to realize its beliefs about what it's sensing you'd have a variational free energy but then you'd also have the um the algorithmic um free energy um um that you could that apply to any partition and the PO if you can show that all all three all three sh share the same minimum at the point of um attaining non equilibrium steady state I think that will be really you know a really nice illustration that all of these are different facets of exactly the same thing and you it is a description of self-organization but articulated in slightly different ways but I repeat you once you've got different algorithms I I think the process of sorting now um um is is somewhat constrained so that it's you because you got three different ways of doing this um they may have different um they may have different functionals that are being minimized um and it may be that but I'm not absolutely sure that the the order matters and as soon as the order matters then you've got um Dynamics in play once you've got Dynamics in play um that that I think slightly complicates the simple um um algorithmic complexity argument because the the algorithmic complexity um the universal computation view um it's not really fit for purpose to understand Dynamic self-organization and indeed most people would argue it's not fit for purpose to to do anything because it's intractable but it's a beautiful mathematical object um does that make sense yeah um uh so so one thing then for with Adam's point so so I think that's a really interesting point and it raises another question which is if if if on the compression so on the compression issue so so if we say that what you're trying to compress is the actual list of numbers plus The Ordering of the algot types then you know everything as as you guys just said but I wonder couldn't somebody argue that in fact there is no list of algot types to compress there's only the numbers because it's sort of like you know by the time you get to the end it's it's it's kind of like um it's immaterial information it doesn't do you know it gets lost by the time you've sorted the numbers what do you need the list of algot types for right they're not really I I don't know I I there's something here no I don't think so like if if you take the position that the algot types aren't relevant once they stop being used then you're imposing as an observer an assumption that the list has finished moving right but like how do you know that yeah yeah no that's that's that's super interesting and and and it's like the bigger question of there is this notion of algot types that maybe you have to take into account what else do you have to take into account that we don't know about right like that's that's one of the things that I see as as so interesting about this and then the next thing I was going to ask you Carl is what's the status of the fact that like all the things that we were just talking about about the cells you know being objects and exchanging information with their neighbors about alot types and having predictions I mean none of that is actually in the algorithm you can see the algorithm is like six lines of quote like you can see what the algorithm is none of that is there so what's the you know it's more of a philosophical question you know what what's the status of of something and and I had the same question when I first heard about you know photons and least action and all that like but there's no mechanism to know you know to calculate which path you know is going to be best for you so do we do with this what what do we do I'm super interested in uh these sort of um I don't know almost almost uh you know uh implicit uh things that it's doing whereas the explicit algorithm doesn't have any of that what what do you think about that um well probably think the same thing that you do um I think um yeah in a sorry in a sense what I was saying about a nuanced answer once you're dealing with your isomorphisms between the local algorithms was exactly this issue that you're bring to the table it's not your um yeah it could be as simple as each algorithm has a different um objective function different free energy or lipun of function um or it could be that they have the same but the um the actual sequence of updates or moves is somehow constrained so the movement on the same free energy surface is is somehow constrainted to be different so I'd have to know precisely what the algorithms are um it probably is the case I'm just guessing um that they probably don't have quite the same um objective function or or and Oran point of view the free energy principle implicit generative model um so the the chimeric um um self-organization is a reflection of the fact that um um not everything is trying to has the same gen model and there for by definition will not have the same free energy um functional so that that does complicate the situation and makes it more interesting in fact um you know from the point of view of this this kind of uh requisite variety um but now i' forgotten the the your your actual question um which I did have an answer to what can you remind me what the sure sure it's so so what these algorithms have in common with some of the things that you and and Chris fields have said about particles and things and and other people of part which is different from what happens in biology right if if if in biology I said look this cell is exchanging information with that cell and it's making decisions the next question is excellent what's the mechanism right like what like show me show me the the the explicit uh set of steps by which the cell does that but but here we don't have that and and presumably you know when we get down to particles and things we don't have that either so what's what what's the status of these all these amazing things that they're doing without a a a mechanism to explicitly do it right um I'm going to give you an answer which um um comes from conversations with philosophers of maths people like Maxell ramstead that that question technically would be answered um by um appeal to what is a mechanics so a mechanics is um for example the basy mechanics of the free energy principle or lran or classical mechanics and uh um certain um Dynamics um uh you know non-dissipative or conservative so or quantum mechanics um where um you can't uh you you you you have to focus exclusively on the on the dissipative uh Dynamics so the mechanics um is a description of the realization of something um where the thing usually conforms to a principle of least action so this is where this is a sort of deflationary answer to your question that the mechanics in and of itself is an emergent property of a variational principle of leased action that can be cast in gauge theoretic terms or in or in terms of things like maximum entropy principles so there are principles um that just describe the shape the SpaceTime shape of our world these um give rise to and usually you know you can usually uh reduce all physics principles to principles of least action the the know the straight line the path of least effort um um and once you've written written down your Principle as a principle um of least action then the particular functional form of the system to which that principle applies then gives you a mechanics and then that now um acquires a teeology in conversation but only in conversation you don't need the mechanic the mechanics does not engineer anything it is just an expression of the principle of least action um so very much in the spirit of basic mechanics I was saying before the free energy things that self-organize you may or may not want to then go on and say oh well this self-organization could be described theologically as self- evidencing or active inference or decision making or basil cognition or distributed intelligence you don't have to do that but it sometimes it can be very useful when talking to somebody else about it to to to te logically frame it like that and that I think is your mission I I think that's what you're bringing to the table in in the widest sense you are saying that the mechanics of biotic self organization have a certain techology which is almost isomorphic to the same Tey of find in Psychiatry or immunotherapy or climate change uh and we just got to find the crosscutting themes so the mechanics the mechanisms um are really just theological unpacking of of of the mechanics so in this instance I gave you an example before that simply um the algorithm uh to implement the algorithm which is probably a series of Boolean operators would need certain inputs they need arguments and they need certain outputs those are the act the sensory active States those those Define now the Markoff blanket um that so you know taking the input is basically whatever my neighbor Neighbors value uh whatever I can see and what I can see is my neighbor's value and it has has to be a neighbor I can't see you know the one for you I can't see somebody a long way away so that defines the input and the output is my particular number that's what I broadcast but I only broadcast it in a Lo in in a local sense so when you starting to express things in terms of a um a teolog of self-organization um of the kind that people use in in you know in the free energy principle I think you're then you're quite licensed to say well this is just a description for example of electrochemical signaling you know there's there's no no magic is this is this is this is the mechanism it just means that there has to be a local signal that um reports or has some um um morphism between my state and um um and you know everything that is not me um and we find multiple instances of this in uh in in biology at different temper and spatial scales and you know the more you drill down the more you actually specify what it is the more mechanistic you it will become but you know at the end of the day that's just the mechanics you're talking about so if you know if you wanted to describe self-organization of massive bodies that had no dissipative aspects to them you know the motion of the planets for example if you if you take yourself can pretend you're Kepler um yeah what is the mechanics of motion of the he Heavenly Bodies they lran conservative mechanics that inherit from the principle of least action that um you know has a relatively simple form before Einstein came along um in terms of uh you know um um energy conservation you know and uh all that is implicit in a path of a path of least you know the least action uh least action principle so that would be his kind of mechanics and you start to invent things like gravity and mass and you know and talk about the tileology of massive bodies being attracted to each other uh and that would be you know quite comfortably received as a an intuitive mechanistic explanation for the thing at hand which in this instan is the motion of heavenly bodies you I I don't see there's any real problem from your point of view you've already told story you've already got the mechanics uh it's just you know a question of um showing how Universal this kind of mechanics is and uh and Universal in in the special context of local interactions um and all the uh all the consequences of having a principle uh or the principles that would apply to self-organizing systems out of equilibrium or non-equilibrium um uh self-organization biotics self organization what kind of mechanics must be in play um and then you can give particular Exemplar and talk about gravity or ell intercellular signaling or um you know and the locality of that does that make sense yeah yeah yeah yeah um and I guess um Adam did you want to say anything before I because I've I've got a little more um another okay so so this is kind of the uh the the really kind of far out thing in and feel free to tell me that this is not a good analogy and you know kind of too too too far out but um I I was thinking I was thinking about the analogy of us trying to analyze these algorithms and what it is that what are the causes of their behavior as we observe them and trying to uh analogize to uh in a in a um you know biological or maybe in a psychological psychiatric context where you have what you think is the algorithm and maybe that's what your subject thinks is the cause of their actions and maybe that's what you think is the cause of their actions but it turns out that there's this underlying Dynamic that is it's it's an extra goal that you didn't know because you couldn't you you didn't see it in the uh in the steps of the of the policies that they're supposedly following and I wonder if this is a sort of really basil kind of I don't know I just just a really basil kind of Behavioral analysis that might be important for uh larger and more complex systems in the sense of finding um underlying goals for for for complex behaviors that are not to be found in the by enumerating the mechanisms that you know that are there which is you know like basically basically looking at the tendency to Cluster uh as a as a hidden motivation for their behavior if I can use this kind of like psychological term what do you think about that is that is that silly or or are there oh no no I I think that's you exactly the application of these principles and the attendant mechanics you know in terms of say vote in Dynamics um or geopolitical um um and um or just a spread of information on the internet you know um I I think some really important deep questions there and and um things you know why is it that almost inevitably whenever you look at some um ideological political or theological commitment everybody's 5050 you know Trump versus Biden brexit versus stay you know wherever you look U the only evolutionary stable strategy on or fre energy minimizing not an equilibrium steady state is you know usually a 5050 but of course that can be subdivided within what this this 50% there's another 50% in a in a sort of self- similar way all the way down so there some must be something generically um very important you diversal about that and I would imagine that's you you will see that kind of clustering it's interesting I didn't realize I I I didn't read it careful enough to realize you had this sort of um that the actual order wins out at the end of the day so you get the increase and then the decrease in clustering that that was interesting it I would be if I was a young man working for as you for you as a PhD student I'd like to put a bit of noise on the numbers and just see if you can keep it alive and dynamic and see if that clustering and that Chic um uh Behavior look at its Dynamics and you know something in dynamical systems theory called frustration that you get in these um um in these chimeric um situations when You' broken detail balance like this which which may be a good metaphor for the voting Dynamics for example uh so I know I think it's a very sensible idea I'm just reminded have you read that paper by Connor Hines um and if not I'll send it to you so he's he's he he um he was making an analogy between um gives energy and free energy um in terms of exchanging ideas um as a model of um Collective behavior um and um it maybe you'll find some interesting ethological references or references to that yeah interesting that you know the thing about the thing about them being pulled apart at the end so so one way one of the things that we we did was to ask okay how how strong would this tendency to Cluster be if you if you didn't have this super overlying you know basic physics of this world that eventually is just going to yank you apart and and one way to do that is to allow repeat numbers so if I allow repeat numbers then you can have a long run of fives and the first half of them could be one algot type the second half could be the other and the actual sorting algorithm would be perfectly happy to keep them as is because the fives are the you know so so we did that and and when you do that you find out that yeah if you let them do that it actually goes higher the the the tendency to Cluster is actually higher that you know so there's this like competing you know and and again looking at it it almost it almost provides a very minimal model of of the existential uh sort of I don't know uh way of life facing living systems right that that the physics of the world are like trying to grind you down but but in the meantime you can do some interesting things that are not not incompatible with them there's no magic I mean it does the algorithm there's no you know there's no errors but but but yet not quite what what the you know what what the end goal is going to be according to the actual physics of the system yeah